

Executive Business Report: Demand Intelligence System

Prepared for: Head of Supply Chain & Chief Financial Officer (CFO)

Prepared by: Aanya Maheshwari

Project Deployment & Source Access

- **GitHub Repository Link:** <https://github.com/Aanya-Maheshwari/Sales-Forecasting-System>
- **Live Streamlit Dashboard Link:** <https://sales-forecasting-system-ham9pqjggmrsapxz4whipg.streamlit.app/>

1. Executive Summary Every retail operation encounters a critical problem: balancing inventory to prevent the heavy capital losses of overstocking while simultaneously avoiding customer dissatisfaction and lost revenue due to understocking. This report outlines an integrated, data-driven Sales Forecasting and Demand Intelligence System built over four years of daily historical superstore sales data. By automating future demand predictions, isolating statistical anomalies, and segmenting product lines based on purchasing behaviors, this system transforms raw transaction data into a reactive tool for weekly operational procurement planning.

2. Core Operational Forecasting (Next 3 Months) Through rigorous testing of three distinct forecasting frameworks, the system successfully modeled upcoming sales trends. Production targets have been established using the most stable model, optimizing procurement buffers across key categories and geographical markets:

- **Upcoming Growth Drivers:** The Technology product category exhibits the strongest upward momentum and growth trajectory for the upcoming quarter, demanding immediate stock prioritization.
- **Peak Regional Marketplace:** The West Region remains our most dominant and expanding sales territory, requiring systematic inventory fulfillment.

Global Model Validation Metrics (Prophet Pipeline):

- **Mean Absolute Error (MAE):** 20,296.01
- **Root Mean Squared Error (RMSE):** 22,487.47
- **Validation MAPE (Error Rate):** 21.89%

3. Historical Sales Outliers & Risk Control Our algorithmic checks identified critical weekly sales anomalies across historical timelines. Rather than systemic errors, these spikes represent predictable retail spikes that supply chain teams must prepare for annually:

1. **Late November Volume Surge:** Directly corresponds to automated pre-holiday and Black Friday promotion periods.
2. **Mid-December Clearance Spike:** Caused by end-of-year inventory liquidation and holiday buying cycles.
3. **Unexpected Sales Drops:** Early quarter corrections where operational purchase volumes naturally cool down.

4. Product Demand Segmentation & Inventory Strategies To avoid a generic, one-size-fits-all inventory framework, product sub-categories have been grouped into three distinct behavior matrixes:

- **High Volume, High Volatility Items:** Drive massive revenue but experience sudden, erratic demand spikes. *Strategy:* Maintain aggressive safety stock buffers to eliminate stockout cycles.
- **Low Volume, Stable Demand Lines:** Highly predictable but slower consumption rates. *Strategy:* Operate on a tight, Just-In-Time (JIT) low-holding baseline replenishment pipeline to avoid freezing working capital.
- **Moderate Volume, Growing Demand:** Items showing upcoming market traction. *Strategy:* Implement milestone-based automated reorder thresholds to scale inventory inline with trend growth.

5. Financial & Supply Chain Recommendations

1. **Capital Optimization:** Transition low-volume stable items to a continuous pipeline replenishment model to free up valuable warehouse floor space and liquid capital.
2. **Buffer Allocation:** Reallocate warehouse safety buffers directly to the Technology category over the next 60 days to capture the upcoming revenue surge without risking out-of-stock events.
3. **Promotion Alignment:** Synchronize procurement schedules with marketing teams 45 days ahead of November to prevent logistic chokepoints during seasonal spikes.

6. System Limitation & Risk Management While highly accurate, this intelligence system relies strictly on historical patterns. It is blind to sudden external shocks such as severe macroeconomic shifts, supplier-side raw material shortages, or sudden competitive price wars. Supply chain leaders should utilize these forecasts as an objective baseline combined with human market insights.